TopoCap Cross-Family Capacitance Transfer

Final close-out report

Concepts, implementation, experiments, results, failures, and next-data requirements

Bottom line

The central objective was not achieved. The study did not demonstrate topology-general transfer learning. It found a small exploratory benefit from support-conditioned source retrieval, but the strongest K = 16 point estimate came from a target-only active-GDS specialist, the fixed v0 transfer path was harmful, and both observed component families had only 3 x 3 capacitance matrices. The reusable output constraints, net-sidecar contract, split controls, and artifact pipeline should be retained; the scientific model claim should not.

Evidence tier: exploratory research archive - not a production recommendation

August 28, 2026

Experiment commit: `d28d180` State digest: `e54eb7d0316726cb...`

Contents

1 Executive summary

1.1 The question

Can a model trained on a large `GeneralizedCapNInterdigital` simulation/GDS corpus reduce the number of labeled simulations needed for a different `CapNInterdigitalTee` component, while remaining compatible with arbitrary capacitance-matrix sizes and layout tools?

1.2 The answer from this study

Not yet. The experiment produced one narrow positive result, several negative or inconclusive results, and one decisive data limitation:

1. **Best K = 16 point estimate:** the target-only active-GDS specialist reached **0.259** macro component log-MAE, better than retrieved-source EBRA (0.293), the evidence-gated system (0.310), target controls (0.330), and v0 transfer (0.404).
2. **Narrow source-label signal:** support-conditioned retrieval improved over its exactly matched shuffled-label control by **+0.0140**, outer-domain 95% interval [**0.0038, 0.0292**], at K = 16. This is exploratory evidence that nearby source labels carried some useful structure.
3. **No broad source-label signal:** full-source canonical controls improved over shuffled labels by only **+0.0005**, interval [**-0.0077, 0.0082**]. This is inconclusive.
4. **v0 was actively unhelpful for this transfer:** its correctly paired labels were **-0.0447** relative to shuffled labels, interval [**-0.0681, -0.0239**]. The negative interval means the compressed v0 path transferred worse than its shuffled-label counterpart.
5. **Topology generalization was not tested in data:** all 20,956 observed matrices had **N = 3**. Tests at N = 2 through 16 verify software properties, not empirical generalization.
6. **Safety was incomplete:** the evidence gate selected transfer in 69% of K = 16 trials, yet the gated system was worse than the target-control baseline in about 37% of trials.
7. **Diffusion was correctly deferred:** the dataset lacks variable-N topologies and repeated alternative geometries for the same electrical target, so a generative inverse-design claim would not be credible.

1.3 Close-out decision

Item	Final disposition
Universal/topology-general ML claim	**Not supported; close this claim**
Cross-family source-label transfer	**Small exploratory retrieval signal only**
Target-native active-GDS descriptors	**Retain; strongest observed representation**
Physical Maxwell output reconstruction	**Retain; correct by construction**
Electrical net sidecar and immutable provenance	**Retain; required infrastructure**
Fixed/compressed v0 as transfer representation	**Retire for this use case**
Evidence gate	**Research prototype only; not a safety guarantee**
Diffusion inverse design	**Do not run until new data and solver-budget controls exist**

2 How this research direction evolved

The work progressed through three increasingly demanding ideas. Each stage exposed a different failure mode.

2.1 Static layout embeddings (v0 and the rejected v1 attempt)

The static v0 representation combined native parameters, geometric moments, and a signed 96 x 96 raster. It was useful for visualization and within-catalogue similarity, but it had no principled electrical-net identity, fixed spatial resolution, and no natural way to compare different capacitance-matrix sizes. An attempted v1 representation was rejected after visual nearest-neighbor and cosine-distribution audits showed worse shape fidelity and excessive similarity crowding relative to v0. It was not retained as a successful standard.

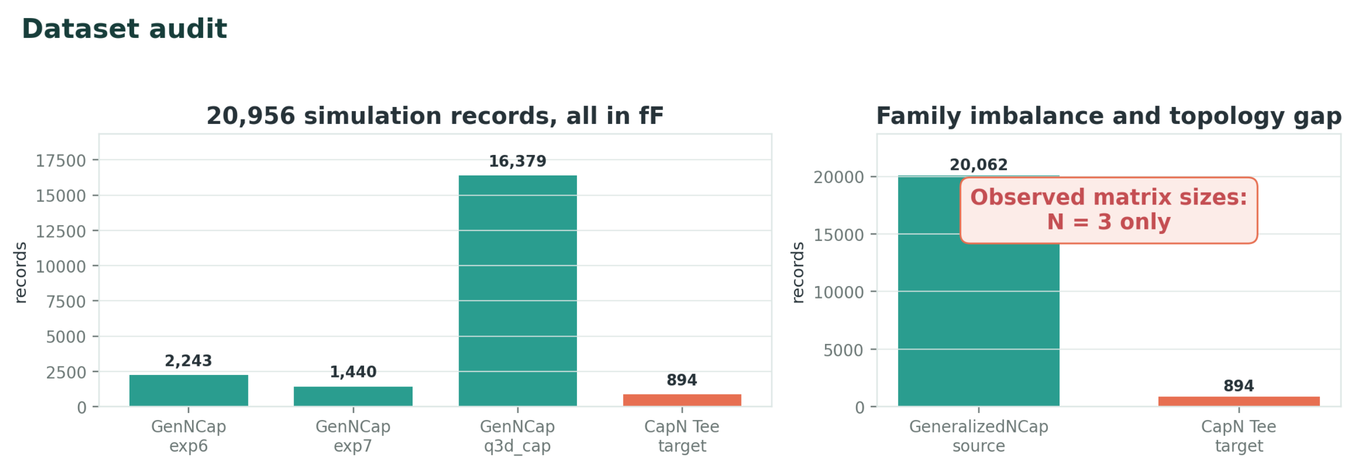

Figure 1: Dataset coverage and topology audit

For this final benchmark, v0 was given a fair compact baseline: all 9,227 coordinates were source-standardized, projected through a target-blind Gaussian sketch to 64 dimensions, then standardized again using source statistics only. The negative source-label result shows that merely compressing the raster did not solve cross-family transfer.

2.2 Within-family finger-count transfer

Tutorials 15, 16, and 16b explored transfer within the GeneralizedNCap family. Those studies were useful for building learning-curve, checkpoint, repeated-split, ablation, and balanced-domain machinery. They did **not** establish cross-component generalization. Early curves were also sensitive to unequal finger-count sample sizes and one-run estimates, which motivated balanced 1,260-record domains and repeated trials. The enduring lesson is methodological: within-family interpolation and random row splits can look excellent while saying little about new geometry families or topology.

2.3 TopoCap-EGRA

TopoCap-EGRA was the attempt to move beyond fixed embeddings. It represented conductors and pairwise interactions explicitly, predicted a physical capacitance matrix, adapted a source model with target support, and used an evidence gate to choose between transfer and a target-native specialist. This report is the final evaluation of that attempt.

3 System concept

3.1 Tool-neutral inputs

Each design was intended to provide three complementary views:

- **GDS-derived geometry:** conductor polygons, areas, perimeters, gaps, overlaps, boundary context, and active-region descriptors;
- **electrical net sidecar:** immutable selectors, ordered net IDs, conductor roles, and the modeled reference;
- **native design controls:** original layout-tool parameter names plus physical role, unit, bounds, constraints, and the path needed to write a prediction back to the generating tool.

The conceptual pipeline was:

GDS + nets + native controls → conductor interaction graph → source foundation / target specialist → evidence gate → physica

3.2 Physical output parameterization

For every conductor pair the model predicts a positive mutual magnitude m_{ij}, and for every conductor a positive shunt s_i. It reconstructs

$$C_{ij} = -m_{ij} \quad (i \neq j), \qquad C_{ii} = s_i + \sum_{j \neq i} m_{ij}.$$

This guarantees symmetry, non-positive off-diagonals, positive diagonals, diagonal dominance, and positive semidefiniteness for any supported N. Permuting node labels only permutes matrix rows and columns. This was one of the strongest engineering outcomes, but **a mathematically variable-size head does not prove learned transfer across sizes**.

3.3 Adaptation and evidence gating

- **EBRA:** a source foundation supplies a prior; a small target support set learns a residual correction with uncertainty.
- **Support-conditioned retrieval:** target support controls select exactly 2,048 source rows using active length, width, and gap. Normalization uses source medians/IQRs only, and source finger-count strata receive equal quotas.
- **Target specialist:** a target-native model uses active-GDS features and no source labels.
- **Evidence gate (EGRA):** cross-validation on support domains chooses among foundation, transfer, and specialist paths. The untouched target test domain is never used in this decision.

4 Experimental design

4.1 Data

Family / campaign	Role	Records	Unique GDS	Matrix size	Unit
GeneralizedNCap / exp6	source	2,243	2,243	3 x 3	fF
GeneralizedNCap / exp7	source	1,440	1,440	3 x 3	fF
GeneralizedNCap / q3d_cap	source	16,379	16,379	3 x 3	fF
CapNInterdigitalTee	target	894	894	3 x 3	fF

There were zero exact cross-family overlaps by design ID, GDS hash, row hash, sidecar hash, or source ID. This prevents literal identity leakage, but the CapN net alignment still depends on legacy generator polygon order; an independently marked asymmetric golden layout remains desirable.

4.2 Splits, budgets, and uncertainty

- Outer evaluation: leave one CapN finger-count domain out for test and use the numerically next domain for validation.
- Independent outer domains: 10 finger-count domains.
- Repeats: 10 nested target-support draws per domain.
- Target label budgets: K = 0, 1, 2, 4, 8, 16, 32, 64, and 128.
- Primary predeclared budget: K = 16.
- Aggregation: average repeats within each domain, then bootstrap the 10 outer domains.
- Evidence tier: exploratory; the protocol was developed while the public target release was available.
- Intervals: pointwise, with no simultaneous multiplicity correction.
- Source normalization: source training rows only.
- Target test rows: never used for fitting, normalization, retrieval, or gate selection.

The compact publication bundle contains 466 learning-curve rows and records 11,200 underlying raw-trial rows in the full run. Raw trials/checkpoints were intentionally omitted from the compact repository bundle, but their hashes are recorded.

4.3 Compared methods

Thirteen paths were evaluated on identical target tests:

1. source-only canonical-control foundation;
2. canonical-control EBRA and its shuffled-source control;
3. support-conditioned 2,048-row retrieval EBRA and its matched shuffled control;
4. source active-GDS EBRA;
5. all cached geometry descriptors EBRA;
6. compressed v0-64 EBRA and its shuffled control;
7. target canonical-control scratch model;
8. target active-GDS specialist;
9. target compressed v0-64 scratch model;
10. evidence-gated TopoCap-EGRA.

4.4 Metrics

- **Macro component log-MAE:** absolute error in log-space shunt/mutual components; lower is better and each design receives equal macro weight.
- **Macro relative Frobenius error:** per-design matrix error divided by the true matrix Frobenius norm; lower is better.
- **Paired gain vs target control:** target-control error minus method error on identical test designs; positive favors the method.
- **Paired gain vs shuffled source:** shuffled-label error minus correct-label error; positive indicates source labels contain useful structure beyond regularization/pipeline effects.
- **Negative-transfer rate:** fraction of paired trials worse than target controls.
- **Physical-valid rate:** fraction satisfying the signed-Maxwell checks.

5 Results

5.1 Learning curves

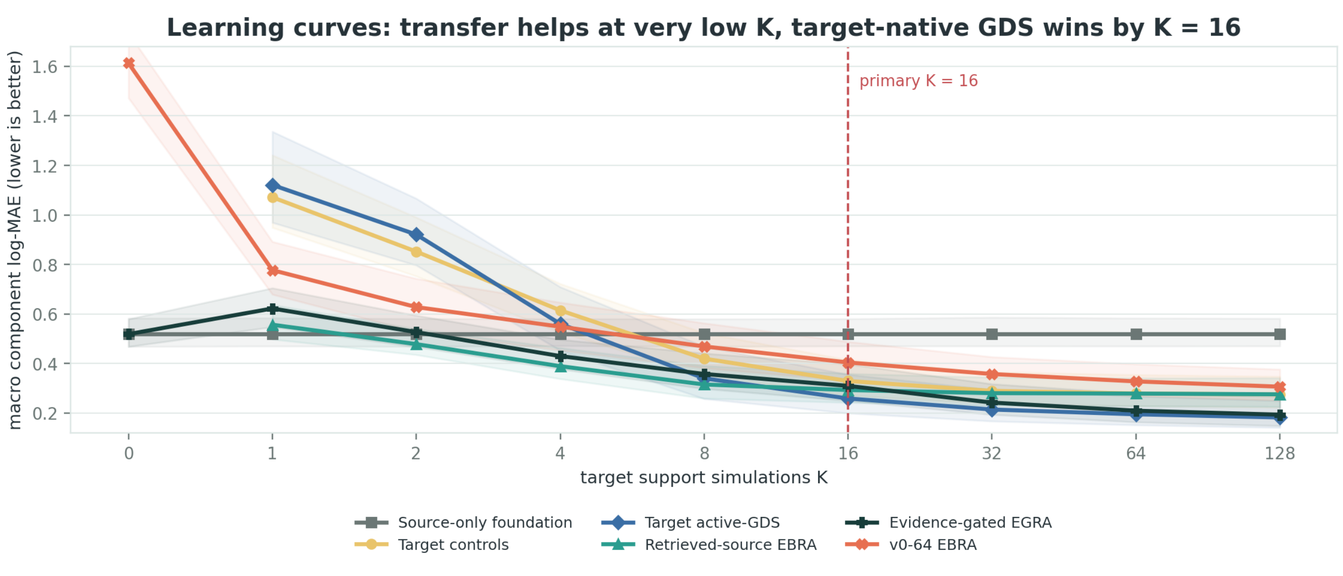

Figure 2: Selected learning curves

The source foundation was poor zero-shot (0.519) and did not improve with target K. Retrieval helped most at very small budgets: at K = 4 it reached 0.389 versus 0.560 for the target active-GDS specialist and 0.615 for target controls. By K = 16 the target active-GDS specialist became best, and remained best through K = 128. The evidence gate increasingly handed off to target-native evidence and reached 0.194 at K = 128, close to the target specialist's 0.182. Retrieval plateaued near 0.276.

This is a useful but narrower finding than the original goal: **source retrieval may help cold-start a new family, but target-native geometry becomes more valuable once a modest support set exists**.

5.2 Primary K = 16 scorecard

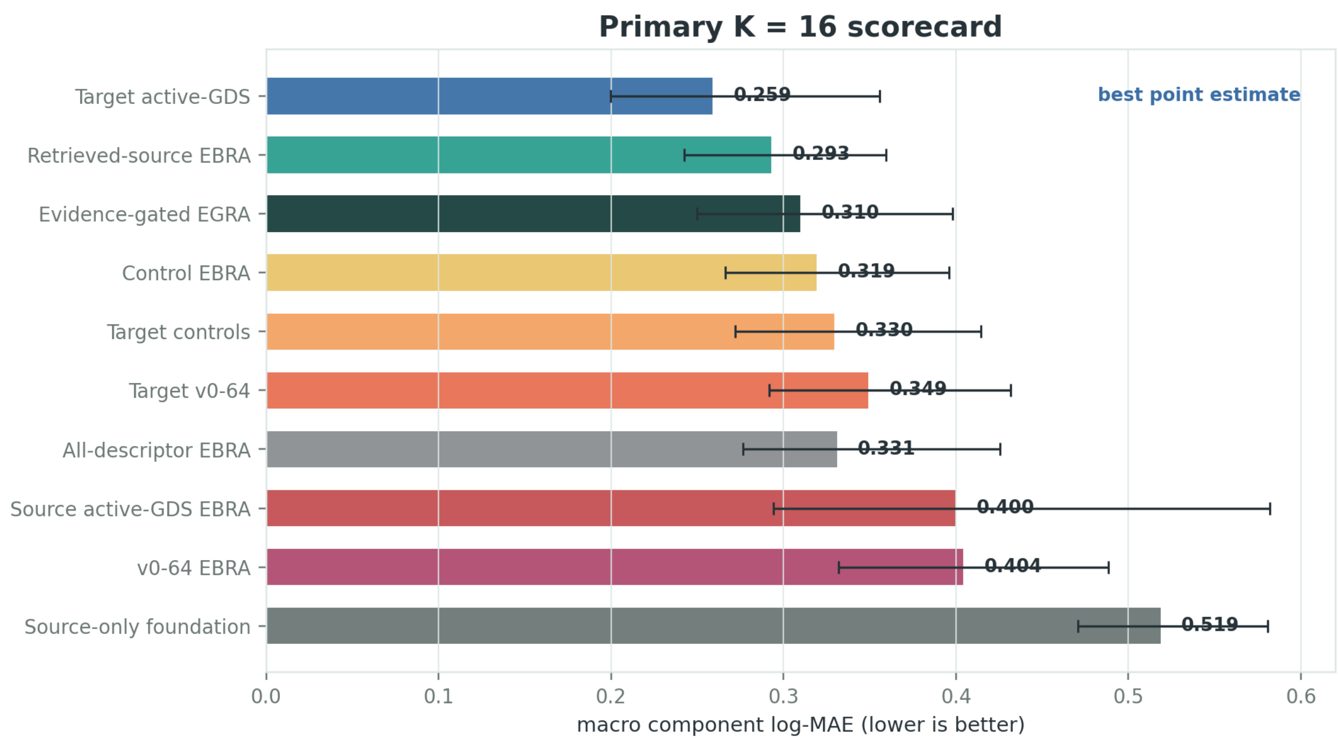

Figure 3: K16 model scorecard

Method	Log-MAE	CI low	CI high
Target active-GDS	0.259	0.200	0.356
Retrieved-source EBRA	0.293	0.242	0.359
Evidence-gated EGRA	0.310	0.250	0.398
Control EBRA	0.319	0.266	0.396
Target controls	0.330	0.272	0.414
Target v0-64	0.349	0.292	0.432
All-descriptor EBRA	0.331	0.276	0.426
Source active-GDS EBRA	0.400	0.294	0.582
v0-64 EBRA	0.404	0.332	0.488
Source-only foundation	0.519	0.471	0.581

The target active-GDS specialist had the lowest loss point estimate (0.259, interval [0.1998, 0.3558]). A ranking of overlapping marginal intervals is descriptive, so claims between methods should use paired contrasts where available. Still, the result directly contradicts any claim that the source foundation was the strongest representation at K = 16.

5.3 What actually carried the gain?

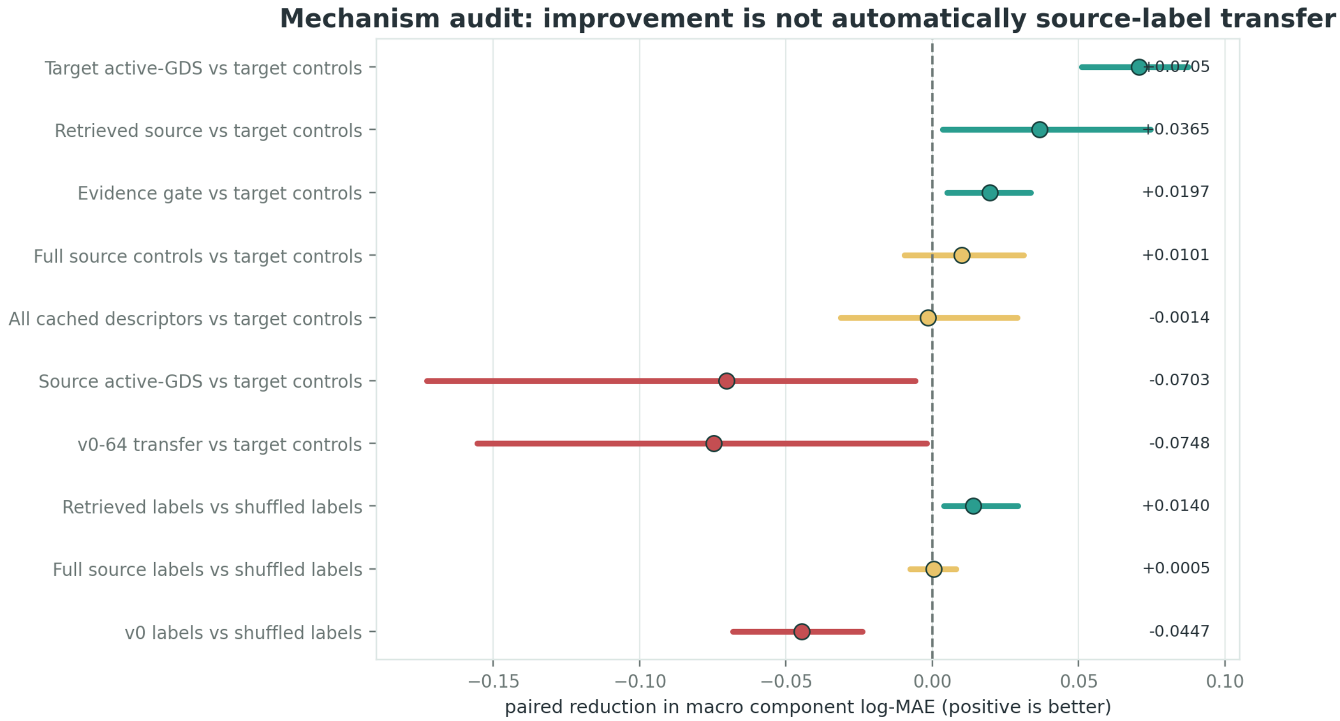

Figure 4: Mechanism forest plot

Paired comparison at K = 16	Estimate	Outer-domain 95% interval	Classification
Target active-GDS vs target controls	+0.0705	[0.0509, 0.0873]	**positive**
Retrieved source vs target controls	+0.0365	[0.0036, 0.0746]	**positive**
Evidence gate vs target controls	+0.0197	[0.0050, 0.0337]	**positive**
Full source labels vs shuffled labels	+0.0005	[-0.0077, 0.0082]	**inconclusive**
Retrieved labels vs matched shuffled labels	+0.0140	[0.0038, 0.0292]	**positive**
v0 labels vs shuffled labels	-0.0447	[-0.0681, -0.0239]	**negative**

Three conclusions follow:

1. **Target-native GDS clearly helped** relative to target parameters alone.
2. **Retrieved source labels carried a small positive signal** relative to a matched shuffled-label pipeline.
3. **The original full-source control path did not establish source-label transfer**, while v0 source labels were harmful. Therefore the overall evidence-gated improvement cannot be described as broad electrostatic knowledge transfer from all 20,062 source labels.

The raw-run sensitivity audit reinforces the caution. The retrieval label effect was positive in 9 of 10 held-out domains and had an exact two-sided sign-flip p-value of 0.0117, but a Student-t 95% sensitivity interval across only 10 domain means crossed zero (approximately -0.0019 to +0.0300). The domain bootstrap interval was positive. This is a promising hypothesis, not a stable universal result.

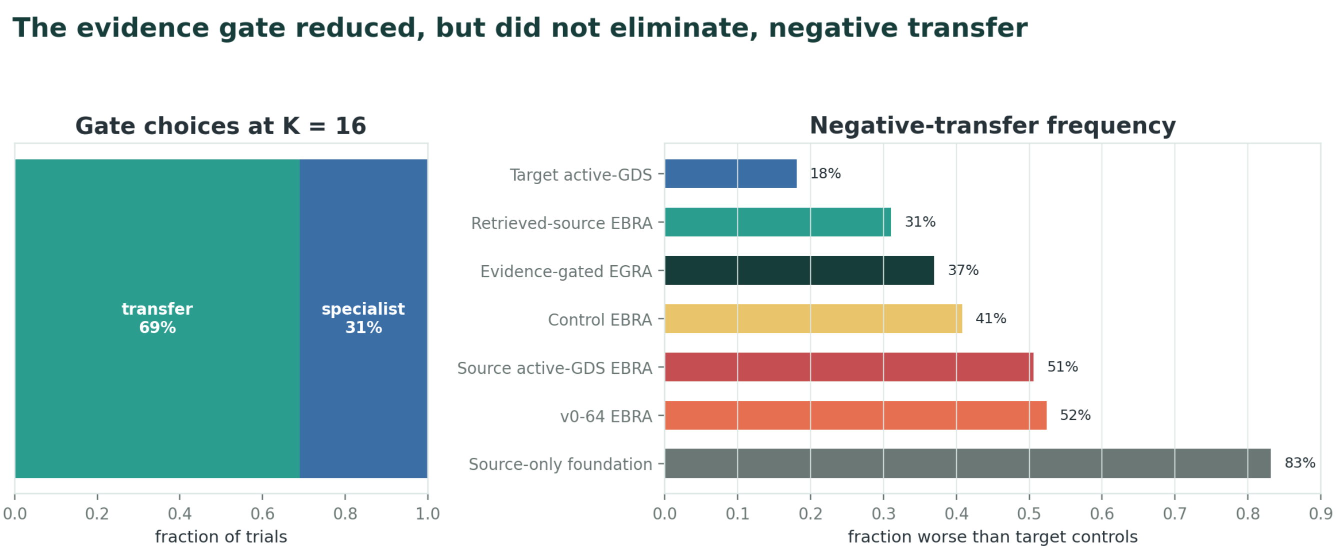

Figure 5: Gate choices and negative transfer

5.4 Evidence gate and negative transfer

At K = 16 the gate selected transfer in 69% of trials, the target specialist in 31%, and the unadapted foundation in 0%. The gated model's paired improvement over target controls was positive (+0.0197, interval [0.0050, 0.0337]), but it still showed negative transfer in about 37% of trials. That is too frequent for the gate to be treated as a production safety mechanism.

5.5 Uncertainty behavior

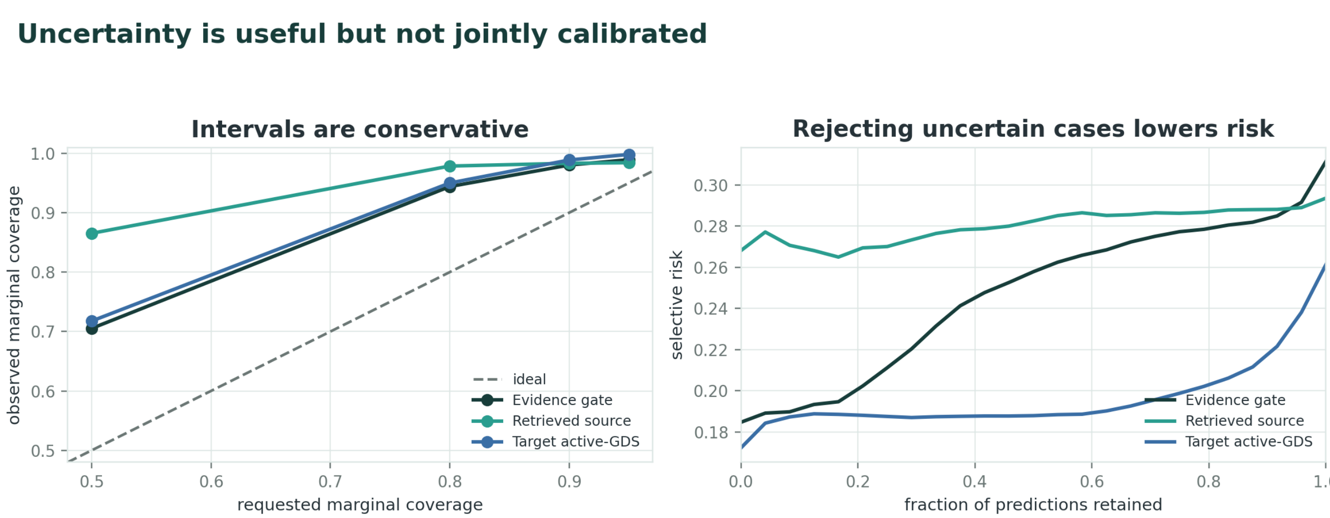

Figure 6: Uncertainty diagnostics

For evidence-gated TopoCap at K = 16:

Requested marginal coverage	Observed coverage
50%	70.5%
80%	94.4%
90%	98.0%
95%	98.9%

The envelopes were conservative, and rejecting the most uncertain predictions reduced error. However, the implementation propagates marginal component intervals into matrix entries. Diagonal entries sum correlated log-normal terms, so these are **not calibrated joint matrix intervals**. A held-out-domain conformal or proper joint posterior calibration layer would be required before using uncertainty as a solver-skipping guarantee.

5.6 Physical and topology properties

All reported methods produced a physical-valid rate of 1.0 because the matrix head enforces the constraints. The architecture/property suite passed all recorded checks:

Property	Tests	Passed	N range	Evidence type
permutation equivariance	15	15	2 to 16	architecture property test
symmetry	15	15	2 to 16	architecture property test
offdiagonal sign violations	15	15	2 to 16	architecture property test
diagonal dominance violation	15	15	2 to 16	architecture property test
psd violation	15	15	2 to 16	architecture property test
component reconstruction	15	15	2 to 16	architecture property test
checkpoint roundtrip	1	1	3 to 3	serialization property test

These results establish implementation invariants and checkpoint reproducibility. They do **not** establish predictive accuracy at N = 2 or N = 4 through 16, because no such matrices occur in the empirical datasets.

6 What worked, what failed, and why

6.1 What worked

1. **Physical output reconstruction:** arbitrary-N signed Maxwell matrices are valid by construction.
2. **Net-aware data contract:** GDS selectors, ordered electrical nets, units, and roles are explicit rather than inferred from bitmap position.
3. **Target active-GDS representation:** the best K = 16 point estimate and strongest paired gain came from target-native active geometry.
4. **Support-conditioned retrieval:** a fixed 2,048-row source subset produced a small positive label-vs-shuffle signal and helped at very low K.
5. **Leakage controls:** source-only normalization, nested support sets, held-out domains, identity-hash audit, matched shuffled labels, and immutable result hashes materially improved scientific reliability.
6. **Reproducibility:** checkpoints, artifact manifests, state digests, and 463 passing repository tests (plus 4 expected live-Hugging-Face skips) make the software result inspectable.

6.2 What failed or remained inconclusive

1. **The universal generalization claim:** only N = 3 was observed; geometry-family transfer cannot substitute for topology transfer.
2. **Zero-shot cross-family prediction:** source-only error was much worse than target-adapted methods.
3. **Broad source-label transfer:** full-source correct-vs-shuffled labels were inconclusive.
4. **v0 transfer:** both target v0 scratch and v0 EBRA underperformed appropriate controls; correct v0 source labels were worse than shuffled labels.
5. **Source active-GDS transfer:** this was significantly worse than target parameters at K = 16, indicating acquisition/cropping/export shift rather than universal geometry semantics.

6. **Evidence-gate safety:** 37% negative-transfer frequency is too high, and the gate's source-label mechanism was not independently established.
7. **Joint uncertainty:** only descriptive marginal envelopes were available.
8. **Confirmation:** the protocol was not preregistered and there were only 10 independent outer domains.

6.3 Why more rows of the same NCap topology will not fix this

The source set is already 22 times larger than the target set, yet source-only and full-source transfer were not dominant. The missing information is not row count; it is variation in **electrical topology, conductor count, process, solver, institution, tool, and geometry family**, together with controlled bridge anchors. Adding another 20,000 N = 3 NCap geometries would mainly improve interpolation inside an already represented manifold.

7 Diffusion and generative design decision

Diffusion remains conceptually relevant for inverse design because many geometries can realize similar capacitance matrices. It was intentionally **not run** here.

A credible diffusion experiment would require:

1. multiple distinct, DRC-valid geometries for overlapping target matrices;
2. explicit electrical nets and variable N;
3. a tool-neutral latent geometry state rather than raw bitmap pixels;
4. tool-specific decoders that round-trip to native parameters and GDS;
5. DRC plus true EM-solver validation;
6. an equal-true-solver-budget comparison against retrieval-seeded trust-region optimization;
7. evaluation of solver success, valid diversity, calibration, and manufacturability.

Until those conditions exist, a diffusion model would mostly learn NCap appearance and produce an attractive but scientifically weak demo.

8 Reusable infrastructure vs retired claims

8.1 Retain

- positive shunt/mutual output decomposition and matrix reconstruction;
- conductor graph and permutation tests;
- explicit `nets.json` contract and signed-Maxwell unit convention;
- native-parameter round-trip metadata;
- immutable GDS/data hashes and checkpoint state digests;
- held-out morphology domains, nested support sets, and shuffled-label controls;
- target-native active-GDS descriptors;
- support-conditioned retrieval as a low-K baseline, not a foundation-model claim.

8.2 Retire or quarantine

- any statement that TopoCap is empirically topology-general;
- compressed v0 as the preferred cross-family transfer representation;
- zero-shot source foundation as a useful predictor for CapN Tee;
- evidence gating as a production-safe transfer mechanism;
- conclusions derived only from random row splits or one training seed;
- diffusion or generative claims before variable-topology solver data exist.

9 Optional next dataset for Saikat Das

This section is a handoff specification, not a recommendation to continue this model immediately. If a future project tests true generalization, collect a **topology bridge dataset**, not another NCap-only sweep.

9.1 One immutable simulation record

```
topology-capacitance-release-v1/
  release.json
  catalogue.parquet
  records/
    <design_id>/
      manifest.json
      layout.gds
      design.json
      nets.json
      process.json
      solver.json
      simulation.json
      raw_solver_export/
```

Required semantics:

- `layout.gds`: final reproducible geometry, top cell, database unit, and SHA-256;
- `design.json`: layout tool/version, component class, full native options, code revision, seed, and for every editable knob its role, unit, legal bounds, constraints, and native option path;
- `nets.json`: ordered node list, modeled reference flag, conductor roles, and immutable GDS selectors;
- `process.json`: layer stack, materials, thicknesses, permittivity, and process ID;
- `solver.json`: solver/version, boundaries, mesh policy, convergence, warnings, runtime, and machine metadata;
- `simulation.json`: raw signed Maxwell matrix in **fF**, explicit node order, solver status, symmetry/sign diagnostics, and raw-export hash.

9.2 Coverage target

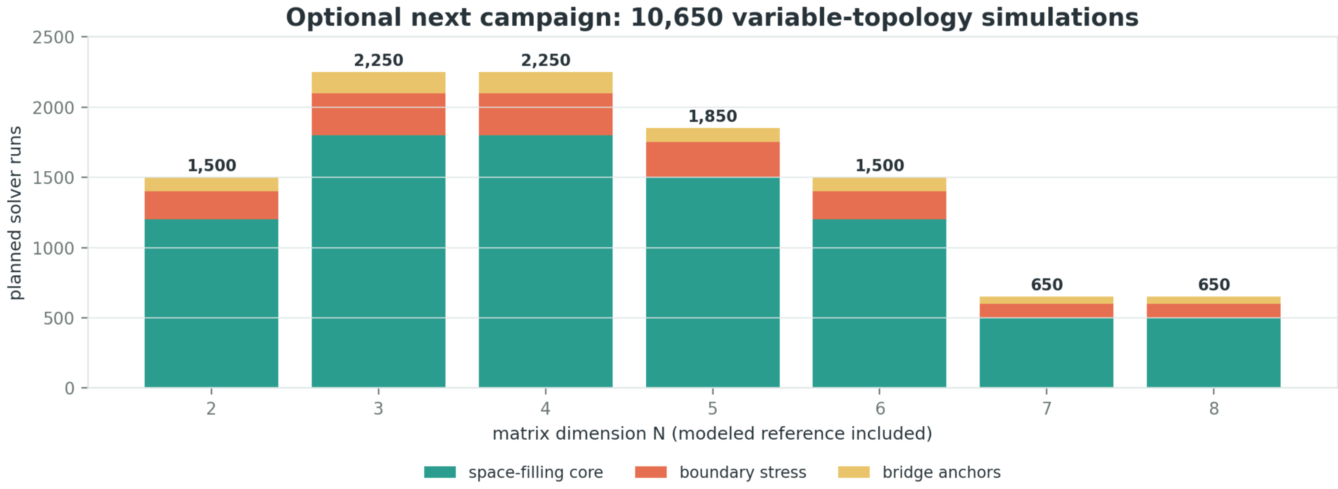

Figure 7: Proposed variable-topology collection budget

N	First geometry families	Core	Stress	Anchors	Total
2	single-ended pads, lines, shielded islands	1,200	200	100	1,500

N	First geometry families	Core	Stress	Anchors	Total
3	non-NCap straight, curved, asymmetric two-signal capacitors	1,800	300	150	2,250
4	three-signal pads, transmon/coupler electrodes	1,800	300	150	2,250
5	resonator-feedline-pad junctions	1,500	250	100	1,850
6	filters, crossovers, electrode arrays	1,200	200	100	1,500
7	scaling challenge	500	100	50	650
8	scaling challenge	500	100	50	650
Total		**8,500**	**1,450**	**700**	**10,650**

Begin with only a **50 to 100 record schema pilot** spanning every planned N, topology family, solver, and process. Validate ingestion and matrix reconstruction before launching the full campaign.

9.3 Sampling strategy

Within each topology/N stratum use Sobol or Latin-hypercube samples over process-relative coordinates:

- conductor width: 1 to 8 times minimum legal width;
- pair gaps: 1 to 20 times minimum legal gap, approximately log-uniform;
- facing/overlap length: 0.05 to 0.9 of active span;
- conductor area ratios: 0.25 to 4 relative to stratum median;
- ground clearance: 1 to 50 times minimum legal gap;
- bend radius: process minimum to 0.25 of active span;
- offsets/rotations across the full DRC-valid range;
- dielectric, metal, solver, and export-path bridge conditions.

Reserve split assignments before simulation. Keep 15% of morphology/topology blocks permanently untouched, dedicate about 10% to weak-coupling and near-DRC stress cases, and include at least 100 byte-identical cross-solver or cross-institution anchors. Re-meshed anchors are repeated measurements, not independent geometries.

9.4 Stop-the-batch QA

Stop and investigate if any of these fail:

- GDS hash does not match the net sidecar;
- any modeled conductor maps to empty or duplicate geometry;
- matrix shape differs from `len(node_order) x len(node_order)`;
- unit or sign convention is missing;
- symmetry residual exceeds the recorded solver tolerance;
- diagonal entries are not positive or off-diagonals are positive beyond tolerance;
- convergence trajectory, warnings, and raw export are missing;
- native options cannot reproduce the GDS hash;

- split group and bridge-anchor IDs were not assigned before model access.

Attribution for the GeneralizedNCap campaign should name **Saikat Das, Levenson-Falk Lab, Eli Levenson-Falk, University of Southern California**, with any solver or fabrication collaborators added per release.

10 The next decisive experiment

If this research direction is revisited, the experiment should be preregistered before the new labels are examined:

1. pretrain on several topology families and N values from one acquisition domain;
2. hold out an entire family plus N, solver, process, tool, or institution combination;
3. adapt on nested K-label support sets from that target domain;
4. evaluate on a separately held-out morphology block;
5. compare target-only parameter, target-only GDS, fixed bitmap, shuffled-source, retrieved-source, and full-target oracle controls at identical solver budgets;
6. report paired domain-level intervals, negative-transfer rates, calibration, physical validity, and true-solver inverse-design success;
7. accept a foundation-model claim only if correct source labels beat matched shuffled labels and target-only baselines across multiple unseen topology/acquisition domains.

The success criterion must be stronger than "the model can emit an N x N matrix." It must show that source simulations lower target data cost on unseen electrical interactions without increasing negative-transfer risk.

11 Reproducibility and provenance

Item	Value
Repository branch	`codex/generalized-transfer-research`
Experiment implementation commit	d28d18090b9a9fab59ef7152e6695ec179b4ea51
Study status	`EXPLORATORY_COMPLETE`
Study-state digest	e54eb7d0316726cb4543919c10a4a0556cd8c8d80b67db63a970de42490140d7
GeneralizedNCap dataset SHA-256	9a13d18fe6c582ebba39e490dddccfbd87d32c68696fd35401098ea7fd6dddb8
CapN target dataset SHA-256	a35365dff70433c7d12404047ec11f48bace1f4962e5c528cdea38bd5ab804d2
Record count	`20,956`
Raw trial rows in full run	`11,200`
Raw-trial SHA-256	688c5880a40b4d22157cbe7c3e7b67e3b7a779cdad40f67b434dfbb228259a4b
v0 standard	`static-embedding-v0`
v0 artifact SHA-256	4511d820fa2fe3fcd90a0d35c0e89e61047fb80aec5af9182d45f5ea2706f6e4
Repository validation	`463 passed, 4 expected live-HF skips; all pre-commit hooks passed`

The compact report data are embedded in this PDF as tables and figures. Canonical machine-readable artifacts remain in `tutorials/data/topocap_transfer_study/`: `manifest.json`, `study-state.json`, `data_audit.csv`, `learning_curves.csv`, `ablations.csv`, `uncertainty.csv`, `topology_checks.csv`, and `diffusion_decision.csv`.

12 Final conclusion

The current TopoCap direction should be closed as an exploratory research archive. It produced sound physical-output and provenance infrastructure, and it identified a plausible low-data retrieval effect. It did **not** produce the requested universally generalizable transfer-learning system.

The most important empirical result is also the simplest: **at $K = 16$, a target-only active-GDS model was better than the source-transfer systems.** The most important scientific limitation is that **both families were $N = 3$**. The most important methodological lesson is that **correct source labels must beat matched shuffled labels before an improvement is called transfer**.

Those conclusions give us a clean stopping point. Future work can start from a different idea without carrying forward an unsupported model claim, while preserving the data contracts and evaluation controls that were genuinely useful.

12.1 Final record snapshot

- **Scientific status:** exploratory and closed; no topology-general claim.
- **Best observed $K = 16$ path:** target-only active-GDS specialist.
- **Only positive source-label mechanism:** modest support-conditioned retrieval effect.
- **Strongest reusable idea:** physical matrix reconstruction plus explicit electrical-net metadata.
- **Required before revisiting:** variable-N, multi-family, multi-acquisition bridge data and a preregistered held-out-domain evaluation.

This document is the complete close-out record for the experiment represented by state digest e54eb7d0316726cb454 3919c10a4a0556cd8c8d80b67db63a970de42490140d7.